Graph Theory 2026 Midterm Test

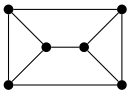
Name _____ Student # _____

- Write neatly, especially your name and student number.
- Clearly $\boxed{\text{box}}$ your final answer if there is one.
- There are 10 questions each worth 5 points, for a total of 50.

1. How many non-labelled connected spanning subgraphs of K_4 are there?

Answer:

2. Determine whether the graph below decomposes into copies of P_4.



3. Let G_n be the graph whose vertices are the permutations of the set $[n]$ and let a permutation $(a_1, \ldots, a_n)$ be adjacent to any permutation that we can get from it by interchanging a_i and a_{i+1} for some i. Show that G is connected.

4. How many copies of C_4 are there in Q_k?

copies

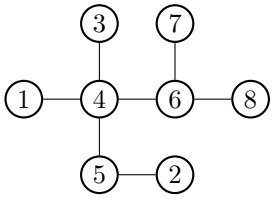
5. Show whether or not the sequence $D = (6, 4, 4, 4, 2, 2, 2)$ is graphic.

6. A tree is a connected acyclic graph. Show that any tree on n vertices has exactly $n - 1$ edges.

7. Show that any edge e of a graph G is in some spanning tree of G.

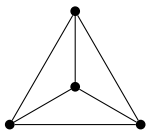
8. Give the Prüffer code of the following tree. (Just the answer is fine.)

Pruffer code



9. Count the spanning trees in the following graph. (You can use whatever technique you like, but be clear about how you did it.)

Spanning trees



10. Use Dijkstra's algorithm on the following graph starting at the vertex r, to find the weighted spanning tree containing the minimum weight rv-path for each vertex v. (Show the order in which you add the edges to the tree.)

